# Hyperbolas

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Standard form**

For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:

**a** the vertices $(\pm 4, 0)$.

**b** the foci $c^2 = 16 + 9 = 25$, so foci $(\pm 5, 0)$.

**c** the equations of the asymptotes $y = \pm\frac{3}{4}x$.

(a) $(\pm 4, 0)$.; (b) $c^2 = 16 + 9 = 25$, so foci $(\pm 5, 0)$.; (c) $y = \pm\frac{3}{4}x$.

**2** **[Foundational]** • **Target: Standard form**

Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.

$a = 2$, $c = 3$, $b^2 = 9 - 4 = 5$: $\frac{y^2}{4} - \frac{x^2}{5} = 1$.

**3** **[Foundational]** • **Target: Standard form**

Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.

**a** Write the equation in standard form by completing the square. $4(x+1)^2 - (y-2)^2 = 4$, i.e. $(x+1)^2 - \frac{(y-2)^2}{4} = 1$.

**b** Identify the conic and state its center. A hyperbola opening left-right with center $(-1, 2)$.

(a) $4(x+1)^2 - (y-2)^2 = 4$, i.e. $(x+1)^2 - \frac{(y-2)^2}{4} = 1$.; (b) A hyperbola opening left-right with center $(-1, 2)$.

**4** **[Foundational]** • **Target: Identifying conics**

Identify each conic section from its equation:

**a** $x^2 + y^2 = 49$ Circle.

**b** $\frac{x^2}{9} + \frac{y^2}{25} = 1$ Ellipse.

**c** $y^2 - 4x^2 = 16$ Hyperbola.

**d** $y = 3x^2 - 6x + 1$ Parabola.

(a) Circle.; (b) Ellipse.; (c) Hyperbola.; (d) Parabola.

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**5** [Foundational] • Target: Identifying conics

A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.

$a = 3$ and $\frac{b}{a} = 2$ gives $b = 6$: $\frac{x^2}{9} - \frac{y^2}{36} = 1$.

**6** [Foundational] • Target: Hyperbolas centered away from the origin

When analyzing the behavior of the model, for the hyperbola $\frac{(y+1)^2}{4} - \frac{(x-3)^2}{9} = 1$, find:

**a** the center $(3, -1)$

**b** the vertices $(3, -1 \pm 2) = (3, 1), (3, -3)$

**c** the direction of opening opens up/down

(a) $(3, -1)$; (b) $(3, -1 \pm 2) = (3, 1), (3, -3)$; (c) opens up/down

**7** [Foundational] • Target: Eccentricity and applications

Find the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, to two decimal places, and state whether it is closer to a pair of straight lines or to a circle.

$c = \sqrt{9 + 16} = 5$: $e = \frac{5}{3} \approx 1.67$. Since $e > 1$ and relatively large, the branches are wide and open quickly, resembling straight lines more than a tight curve.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Eccentricity and applications

A ship uses two radio towers 200 km apart (foci) to navigate via hyperbolic positioning: the difference in distances from the ship to each tower is constant at 120 km. Write the equation of the hyperbola (centered at the midpoint of the towers, opening left-right), in standard form.

$2a = 120$, so $a = 60$; $2c = 200$, so $c = 100$; $b^2 = c^2 - a^2 = 10000 - 3600 = 6400$. Equation: $\frac{x^2}{3600} - \frac{y^2}{6400} = 1$.

**9** [Intermediate] • Target: Graphing a hyperbola branch

The upper-right branch of $y^2 - x^2 = 4$ (as $y = \sqrt{4 + x^2}$) is shown. Use the graph to estimate $y$ at $x = 3$, then verify algebraically, and compare to the asymptote $y = x$.

From the graph, $y \approx 3.6$ at $x = 3$. Algebraically: $y = \sqrt{4 + 9} = \sqrt{13} \approx 3.61$, which is close to but above the asymptote value $y = x = 3$, consistent with the branch approaching but never touching its asymptote.

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**10** **[Intermediate]** • **Target: The hyperbola's defining property**

State the geometric (difference-of-distances) definition of a hyperbola, and use it to verify the vertex $(4, 0)$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ (foci $(\pm 5, 0)$) satisfies it.

A hyperbola is the set of points where the difference of distances to the two foci is constant ($= 2a$). At $(4, 0)$: distance to $(5, 0)$ is $1$; distance to $(-5, 0)$ is $9$. Difference $= 9 - 1 = 8 = 2a = 2(4)$ ✓.

**11** **[Intermediate]** • **Target: Comparing all four conic sections**

State the key algebraic feature (coefficient signs) that distinguishes a circle, ellipse, parabola, and hyperbola in a general second-degree equation $Ax^2 + By^2 + \ldots = 0$.

Circle: $A = B$ (same sign, equal). Ellipse: $A, B$ same sign, unequal. Parabola: exactly one of $A, B$ is zero (only one variable squared). Hyperbola: $A, B$ opposite signs.

**12** **[Intermediate]** • **Target: Finding the equation from asymptotes and a point**

A hyperbola centered at the origin has asymptotes $y = \pm\frac{2}{3}x$ and passes through $(3, 0)$. Find its equation.

Since it passes through $(3, 0)$, it opens left-right with $a = 3$. From $\frac{b}{a} = \frac{2}{3}$: $b = 2$. Equation: $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

**13** **[Intermediate]** • **Target: Applications: LORAN navigation**

Two LORAN radio towers are 260 km apart. A ship determines the difference in distances to the two towers is 100 km. Write the equation of the hyperbola of possible ship locations (centered at the midpoint, opening along the line connecting the towers).

$2a = 100 \Rightarrow a = 50$; $2c = 260 \Rightarrow c = 130$; $b^2 = c^2 - a^2 = 16900 - 2500 = 14400$. Equation: $\frac{x^2}{2500} - \frac{y^2}{14400} = 1$.

**14** **[Intermediate]** • **Target: Graphing a hyperbola branch**

Write the general circle equation $x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, identify its center and radius, and find the equation of the tangent line to the circle at the point $P(4, 1)$.

Complete the square for $x$ and $y$: $(x^2 - 8x + 16) + (y^2 + 6y + 9) = -9 + 16 + 9 = 16$.
$(x - 4)^2 + (y + 3)^2 = 16$. Center is $C(4, -3)$ and radius is $r = \sqrt{16} = 4$. Check the point $P(4, 1)$: $(4 - 4)^2 + (1 + 3)^2 = 0 + 16 = 16$ ✓. The radius connects $C(4, -3)$ to $P(4, 1)$, which is a vertical line $x = 4$. Therefore, the tangent line through $P(4, 1)$ is horizontal: $y = 1$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • Target: The hyperbola's defining property • Cross-Lesson: 9.03 The Binomial Theorem

Write $9x^2 + 4y^2 - 36x + 8y + 4 = 0$ in standard form and identify the conic.

Complete the square: $9(x-2)^2 + 4(y+1)^2 = 36 \Rightarrow \frac{(x-2)^2}{4} + \frac{(y+1)^2}{9} = 1$. Ellipse, centre $(2,\ -1)$, $a = 3$, $b = 2$.

**16** [Challenging] • Target: Comparing all four conic sections • Cross-Lesson: 1.01 Functions and Function Notation

Determine the type of conic section represented by the general quadratic equation $4x^2 - 9y^2 + 16x + 54y - 101 = 0$, and find the coordinates of its center.

Step 1: Identify conic type: The coefficients of $x^2$ $(+4)$ and $y^2$ $(-9)$ have opposite signs ( $A \cdot C = 4(-9) = -36 hyperbola. Step 2: Complete the square to locate center: 4(x^2 + 4x + 4) - 9(y^2 - 6y + 9) = 101 + 4(4) - 9(9) 4(x + 2)^2 - 9(y - 3)^2 = 101 + 16 - 81 = 36 Divide through by 36: \frac{(x + 2)^2}{9} - \frac{(y - 3)^2}{4} = 1. x$.

**17** [Challenging] • Target: Finding the equation from asymptotes and a point • Cross-Lesson: 1.02 Domain and Range

Two chords $AB$ and $CD$ of a circle intersect at interior point $M$. If $AM = 4$ cm, $MB = 9$ cm, and $CM = 6$ cm, determine the length of $MD$.

Step 1: Apply the Intersecting Chords Theorem: $AM \cdot MB = CM \cdot MD$. Step 2: Substitute given lengths: $4 \times 9 = 6 \times MD \Rightarrow 36 = 6 \cdot MD$. Step 3: Solve for $MD$: $MD = \frac{36}{6} = \mathbf{6}$ cm.

**18** [Challenging] • Target: Applications: LORAN navigation • Cross-Lesson: 1.03 Average Rate of Change

Find the constant term (independent of $x$) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

Step 1: Write the general term $T_{k+1}$: $T_{k+1} = \binom{9}{k}(2x^2)^{9-k}\left(-\frac{1}{x}\right)^k = \binom{9}{k}2^{9-k}x^{2(9-k)}(-1)^k x^{-k} = \binom{9}{k}2^{9-k}(-1)^k x^{18-3k}$. Step 2: Find $k$ for the term independent of $x$: Set exponent equal to 0: $18 - 3k = 0 \Rightarrow 3k = 18 \Rightarrow k = 6$. Step 3: Evaluate coefficient at $k = 6$: $\binom{9}{6} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$. .

**19** [Challenging] • Target: Standard form • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A polynomial $f(x)$ of degree 4 with real coefficients has leading coefficient 2 and zeros at $x = -1$ (multiplicity 2) and $x = 2 + i$. Write $f(x)$ in fully expanded standard form.

Step 1: Apply the Conjugate Root Theorem: Since $f(x)$ has real coefficients, the complex conjugate $2 - i$ must also be a zero. Step 2: Construct the factored form: $f(x) = 2(x+1)^2((x-2)-i)((x-2)+i) = 2(x^2+2x+1)((x-2)^2+1)$. Step 3: Expand and simplify: $(x-2)^2 + 1 = x^2 - 4x + 5$.

$f(x) = 2(x^2+2x+1)(x^2-4x+5) = 2(x^4-2x^3-2x^2+6x+5) = \mathbf{2x^4 - 4x^3 - 4x^2 + 12x + 10}$.

**20** [Challenging] • Target: Identifying conics • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$ .

Step 1: Identify progression parameters: First term $a_1 = 4$, common difference $d = 9 - 4 = 5$. Step 2: Compute $a_{15}$: $a_{15} = a_1 + (15-1)d = 4 + 14(5) = 4 + 70 = 74$. Step 3: Compute finite sum $S_{20}$:

$S_{20} = \frac{20}{2}[2a_1 + (20-1)d] = 10[2(4) + 19(5)] = 10[8 + 95] = 10(103) = 1030$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Standard form

Find the equation of the hyperbola with asymptotes $y = \pm 2x$ and passing through $(3, 4)$.

**a.** $\frac{x^2}{5} - \frac{y^2}{20} = 1$ ✓

**b.** $y^2 - 4x^2 = 0$

**c.** $4x^2 - y^2 = 32$

**d.** $x^2 - 4y^2 = 4$

A hyperbola with asymptotes $y = \pm 2x$ has the form $4x^2 - y^2 = k$. Substituting $(3, 4)$ gives $4(9) - 16 = 20$, so $4x^2 - y^2 = 20 \Rightarrow \frac{x^2}{5} - \frac{y^2}{20} = 1$.

**22** [Foundational] • Target: Identifying conics

A hyperbola has vertices $(\pm 2, 0)$ and asymptotes $y = \pm 3x$. Find its equation.

**a.** $\frac{x^2}{4} - \frac{y^2}{9} = 1$

**b.** $\frac{x^2}{4} - \frac{y^2}{36} = 1$ ✓

**c.** $y^2/9 - x^2/4 = 1$

**d.** $\frac{x^2}{9} - \frac{y^2}{4} = 1$

$a = 2$, slope $= b/a = 3 \Rightarrow b = 6$. $x^2/4 - y^2/36 = 1$.

**23** [Foundational] • Target: Hyperbolas centered away from the origin

A hyperbola is defined by the equation $\frac{(x-h)^2}{16} - \frac{(y-k)^2}{9} = 1$. If the vertices of the hyperbola are located at $(5, 2)$ and $(-1, 2)$, determine the values of $h$ and $k$.

**a.** **h = 2, k = 2** ✓

**b.** h = 2, k = -1

**c.** h = 2, k = 5

**d.** h = 5, k = 2

The standard form of a horizontal hyperbola centered at $(h, k)$ is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$. The vertices of such a hyperbola are located at $(h \pm a, k)$. Given the vertices are $(5, 2)$ and $(-1, 2)$, we can equate these to $(h + a, k)$ and $(h - a, k)$ respectively. From the y-coordinates of the vertices, we see that $k = 2$. From the x-coordinates, we have two equations: 1) $h + a = 5$ 2) $h - a = -1$ Adding these two equations together, we get $(h + a) + (h - a) = 5 + (-1)$, which simplifies to.

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**24** [Foundational] • Target: Eccentricity and applications

Prove that the difference of the focal radii of a point on a hyperbola equals $\pm 2a$.

**a. From definition $|PF_1 - PF_2| = 2a$, which defines the hyperbola ✓**

**b.** Only for $e > 2$

**c.** $PF_1 - PF_2 = 2c$

**d.** It equals $2b$

Applying Conic Sections principles, Definition of hyperbola. Therefore, the verified correct option is From definition $|PF_1 - PF_2| = 2a$, which defines the hyperbola.

**25** [Foundational] • Target: Graphing a hyperbola branch

Write the equation of the hyperbola with foci $(\pm 5, 0)$ and vertices $(\pm 3, 0)$.

**a.** $\frac{x^2}{9} - \frac{y^2}{16} = 1$ ✓

**b.** $x^2 - y^2 = 9$

**c.** $\frac{x^2}{16} - \frac{y^2}{9} = 1$

**d.** $\frac{x^2}{25} - \frac{y^2}{9} = 1$

Applying Conic Sections principles, $a = 3, c = 5, b^2 = 25 - 9 = 16$. Therefore, the verified correct option is $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

**26** [Foundational] • Target: The hyperbola's defining property

The eccentricity of a hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ is:

**a.** $\frac{4}{3}$

**b.** $\frac{5}{3}$ ✓

**c.** $\frac{3}{4}$

**d.** $\frac{3}{5}$

Applying Conic Sections principles, $c = \sqrt{9 + 16} = 5$. $e = c/a = 5/3$. Therefore, the verified correct option is $\frac{5}{3}$.

**27** [Foundational] • Target: Comparing all four conic sections

Find the equation of the circle centered at $(0, 0)$ passing through $(3, 4)$.

**a.** $x^2 + y^2 = 7$

**b.** $x^2 + y^2 = 5$

**c.** $x^2 + y^2 = 12$

**d.** $x^2 + y^2 = 25$ ✓

Applying Conic Sections principles, $r = \sqrt{9 + 16} = 5$. $x^2 + y^2 = 25$. Therefore, the verified correct option is $x^2 + y^2 = 25$.

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**28** [Foundational] • Target: Finding the equation from asymptotes and a point

For a hyperbola, the eccentricity satisfies:

**a.** $e > 1$ ✓

**b.** $e = 1$

**c.** $0 < e < 1$

**d.** $e = 0$

Applying Conic Sections principles, Hyperbola: $e > 1$. Therefore, the verified correct option is $e > 1$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Applications: LORAN navigation

Find the angle between the asymptotes of the hyperbola $\frac{x^2}{4} - \frac{y^2}{4} = 1$.

**a.** $45^\circ$

**b.** $90^\circ$ ✓

**c.** $60^\circ$

**d.** $30^\circ$

$a = b$: rectangular hyperbola. Asymptotes $y = \pm x$: angle $90^\circ$.

**30** [Intermediate] • Target: Standard form

The rectangular hyperbola $xy = c^2$ has eccentricity:

**a.** 1

**b.** 2

**c.** $\sqrt{2}$ ✓

**d.** $\sqrt{3}$

For $x^2/a^2 - y^2/b^2 = 1$ with $a = b$: $e = \sqrt{2}$.

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**31** [Intermediate] • Target: Identifying conics

Find the equation of a sphere in $\mathbb{R}^3$ centered at $(1, -2, 2)$ with radius $3$.

**a.** $x^2 + y^2 + z^2 = 9$

**b.** $(x-1)^2 + (y+2)^2 + (z-2)^2 = 3$

**c.** $(x+1)^2 + (y-2)^2 + (z+2)^2 = 9$

**d.** $(x-1)^2 + (y+2)^2 + (z-2)^2 = 9$ ✓

$(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2 = r^2$.

**32** [Intermediate] • Target: Hyperbolas centered away from the origin

Find the equation of the hyperbola with transverse axis along $y$, $a = 3$, $b = 4$.

**a.** $\frac{y^2}{16} - \frac{x^2}{9} = 1$

**b.** $\frac{x^2}{9} - \frac{y^2}{16} = 1$

**c.** $\frac{y^2}{9} - \frac{x^2}{16} = 1$ ✓

**d.** $y^2 - x^2 = 3$

Transverse axis along $y$: $y^2/a^2 - x^2/b^2 = 1$.

**33** [Intermediate] • Target: Eccentricity and applications

Find the equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $(a\cos\theta, b\sin\theta)$.

**a.** $x = a\cos\theta$

**b.** $x\cos\theta/a + y\sin\theta/b = 1$

**c.** Normal cannot be found in closed form

**d.** $ax\sec\theta - by\csc\theta = a^2 - b^2$ ✓

Equation of normal at parametric point.

**34** [Intermediate] • Target: Graphing a hyperbola branch

A hyperbola has $a = 4$, $b = 3$. Find the equations of the asymptotes.

**a.** $y = \pm 4x$

**b.** $y = \pm 3x$

**c.** $y = \pm\frac{3}{4}x$ ✓

**d.** $y = \pm\frac{4}{3}x$

Asymptotes: $y = \pm(b/a)x = \pm(3/4)x$.

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**35** [Intermediate] • Target: The hyperbola's defining property

Find the equation of the circle with center $(3, -1)$ and radius 5.

**a.** $(x+3)^2+(y-1)^2=25$

**b.** $(x-3)^2+(y+1)^2=25$ ✓

**c.** $(x-3)^2+(y-1)^2=5$

**d.** $x^2+y^2=25$

Applying Conic Sections principles, Standard circle equation. Therefore, the verified correct option is $(x-3)^2+(y+1)^2=25$.

**36** [Intermediate] • Target: Comparing all four conic sections

The equation of an ellipse is given by: $9x^2+4y^2-36x+8y+k=0$ If the length of the major axis of the ellipse is $6$, what is the value of the constant $k$?

**a.** $4$ ✓

**b.** $-31$

**c.** $-41$

**d.** $-104$

To determine $k$, we rewrite the equation in standard form by completing the square for both $x$ and $y$: 1. Group the $x$ and $y$ terms and isolate the constant $k$: $9(x^2-4x)+4(y^2+2y)=-k$ 2. Complete the square inside each set of parentheses. Remember to distribute the leading coefficients when adding terms to the right-hand side: $9(x^2-4x+4)+4(y^2+2y+1)=-k+9(4)+4(1)$ $9(x-2)^2+4(y+1)^2=40-k$ 3. Divide both sides by $40-k$ (assuming $40-k>0$).

**37** [Intermediate] • Target: Finding the equation from asymptotes and a point

An ellipse in the $xy$-plane is defined by the equation $9x^2+16y^2-54x+64y+c=0$ where $c$ is a constant. If the length of the minor axis of the ellipse is $6$, what is the value of $c$?

**a.** $1$ ✓

**b.** $-1$

**c.** $64$

**d.** $289$

To find $c$, convert the given equation into standard form by completing the square for both $x$ and $y$: 1. Group the $x$ and $y$ terms: $(9x^2-54x)+(16y^2+64y)=-c$ 2. Factor out the leading coefficients: $9(x^2-6x)+16(y^2+4y)=-c$ 3. Complete the square inside each set of parentheses: $9(x^2-6x+9)+16(y^2+4y+4)=-c+9(9)+16(4)$ $9(x-3)^2+16(y+2)^2=145-c$ 4. Divide by $145-c$ to write the equation in standard form.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Applications: LORAN navigation • Cross-Lesson: 1.03 Average Rate of Change

The function $f$ is defined by $f(x) = \frac{2x^2 + kx - 6}{x - 2}$ for all real numbers $xeq2$, where $k$ is a constant. If the graph of $y = f(x)$ in the standard $(x, y)$ coordinate plane has a removable discontinuity (a hole) at the point $(2, L)$, what is the value of $L$?

**a.** $-1$

**b.** $3$

**c.** $7$ ✓

**d.** $11$

A rational function has a removable discontinuity at $x = 2$ if and only if both the numerator and the denominator share the factor $(x - 2)$. Therefore, the numerator must equal 0 when $x = 2$: $2(2)^2 + k(2) - 6 = 0\ 8 + 2k - 6 = 0\ 2k + 2 = 0 \Rightarrow k = -1$ Now, substitute $k = -1$ back into the numerator and factor it: $2x^2 - x - 6 = (x - 2)(2x + 3)$ For all $xeq2$, simplify $f(x)$: $f(x) = \frac{(x - 2)(2x + 3)}{x - 2} = 2x + 3$ The $y$-coordinate $L$ of the removable discontinuity is the limit of.

**39** [Challenging] • Target: Standard form • Cross-Lesson: 2.01 Right-Triangle Trigonometry

In the diagram below, chords AB and CD intersect at point P. If AP = 6, PB = 8, and CP = 4, find the length of PD.

**a.** **3** ✓

**b.** 4.5

**c.** 6

**d.** 12

The intersecting chords theorem states that when two chords intersect inside a circle, the product of the lengths of the segments of one chord is equal to the product of the lengths of the segments of the other chord. Let the chords be AB and CD, intersecting at point P. According to the theorem, we have: AP × PB = CP × PD We are given the following lengths: AP = 6 PB = 8 CP = 4 We need to find the length of PD. Substitute the given values into the equation: 6 × 8 = 4 × PD 48 = 4.

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**40** [Challenging] • Target: Identifying conics • Cross-Lesson: 2.02 Radian Measure and Arc Length

What are the coordinates of the center and the radius of the circle given by the equation $(x-4)^2+(y+1)^2=36$?

**a.** **Center $(4,-1)$; radius $6$** ✓

**b.** Center $(-4,1)$; radius $6$

**c.** Center $(4,-1)$; radius $36$

**d.** Center $(-4,1)$; radius $36$

The standard form of the equation of a circle is $(x-h)^2+(y-k)^2=r^2$, where $(h,k)$ is the center and $r$ is the radius. Matching the given equation $(x-4)^2+(y+1)^2=36$ to the standard form: 1. $x-h=x-4\Rightarrow h=4$ 2. $y-k=y-(-1)\Rightarrow k=-1$ 3. $r^2=36\Rightarrow r=\sqrt{36}=6$ Therefore, the center is $(4,-1)$ and the radius is $6$. Common misconceptions: - Forgetting the negative signs in $(x-h)$ and $(y-k)$ leads to the incorrect center $(-4,1)$. - Forg.

**41** [Challenging] • Target: Hyperbolas centered away from the origin • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

What are the center and radius of the circle given by the equation $(x-2)^2+(y+6)^2=36$?

**a.** **Center $(2,-6)$, radius $6$** ✓

**b.** Center $(-2,6)$, radius $6$

**c.** Center $(2,-6)$, radius $36$

**d.** Center $(-2,6)$, radius $36$

The standard form for the equation of a circle is: $(x-h)^2+(y-k)^2=r^2$ where $(h,k)$ is the center of the circle and $r$ is the radius. Comparing $(x-2)^2+(y+6)^2=36$ to standard form: 1. $x-h=x-2\Rightarrow h=2$ 2. $y-k=y-(-6)\Rightarrow k=-6$ 3. $r^2=36\Rightarrow r=\sqrt{36}=6$ Thus, the center is $(2,-6)$ and the radius is $6$. Common errors: - Reversing the signs of the coordinates leads to $(-2,6)$. - Forgetting to take the square root of the constant term leads t.

**42** [Challenging] • Target: Eccentricity and applications • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

What is the center of the circle represented by the equation $(x-3)^2+(y+5)^2=16$?

**a.** (3, 5)

**b.** **(3, -5)** ✓

**c.** (-3, 5)

**d.** (-3, -5)

The standard form of a circle's equation is $(x-h)^2+(y-k)^2=r^2$, where $(h,k)$ is the center of the circle. In the given equation, $(x-3)^2+(y+5)^2=16$, we can identify $h=3$ and $k=-5$. Therefore, the center of the circle is $(3,-5)$.

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**43** [Challenging] • Target: Graphing a hyperbola branch • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Prove that the chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ on $y^2 = 4ax$ has equation $y(t_1 + t_2) = 2x + 2at_1t_2$.

**a.** Use calculus

**b.** Incorrect formula

**c.** **Slope $= 2/(t_1 + t_2)$; using point-slope through $(at_1^2, 2at_1)$ gives the result ✓**

**d.** Only for $t_1 = t_2$

Applying Conic Sections principles, Algebraic derivation. Therefore, the verified correct option is Slope $= 2/(t_1 + t_2)$; using point-slope through $(at_1^2, 2at_1)$ gives the result.

**44** [Challenging] • Target: The hyperbola's defining property • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Derive the eccentricity of a conic from the focus-directrix ratio definition $PF/PD = e$ for the parabola.

**a.** $e = 2$ for parabola

**b.** **For parabola, $e = 1$: distance to focus equals distance to directrix ✓**

**c.** $e > 1$ for parabola

**d.** $e < 1$ always

Applying Conic Sections principles, Definition of parabola. Therefore, the verified correct option is For parabola, $e = 1$: distance to focus equals distance to directrix.

**45** [Challenging] • Target: Comparing all four conic sections • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the equation of the ellipse with foci $(\pm 3, 0)$ and sum of focal radii $= 10$.

**a.** $\frac{x^2}{10} + \frac{y^2}{9} = 1$

**b.** $\frac{x^2}{9} + \frac{y^2}{16} = 1$

**c.** $\frac{x^2}{25} + \frac{y^2}{16} = 1$ ✓

**d.** $x^2/16 + y^2/9 = 1$

Applying Conic Sections principles, $2a = 10$, $a = 5$. $b^2 = 25 - 9 = 16$. Therefore, the verified correct option is $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

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